

DUSHANBE, Tajikistan

WMO: 388360

Lat: 38.583N

Lon: 68.733E

Elev: 800

StdP: 92.08

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-5.9	-13.1	1.3	-2.0	-11.0	1.6	1.0	7.4	8.8	6.7	9.3	1.1	270	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	38.2	19.4	37.1	18.8	35.9	18.5	22.5	33.7	21.2	32.7	20.2	31.8	2.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	15.1	28.8	17.4	13.7	27.1	16.1	12.7	25.4	70.6	33.8	65.7	32.6	61.9	32.1	33.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.6	4.8	-11.5	40.6	3.9	1.0	-14.4	41.3	-16.7	41.9	-18.9	42.5	-21.7	43.3		
(5)			-11.7	24.2	3.9	1.2	-14.6	25.1	-16.8	25.8	-19.0	26.5	-21.9	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	3.9	5.3	11.6	16.0	20.6	25.2	27.5	26.2	21.8	16.1	10.1	4.8	(6)
(7)		DBStd	8.99	4.36	5.05	4.32	4.06	3.01	2.47	1.89	1.98	2.53	3.41	4.17	4.13	(7)
(8)		HDD10.0	584	192	139	34	6	0	0	0	0	0	2	46	164	(8)
(9)		HDD18.3	1881	449	364	211	88	13	0	0	0	4	85	248	418	(9)
(10)		CDD10.0	2709	2	9	83	187	327	457	543	503	355	191	49	4	(10)
(11)		CDD18.3	964	0	0	2	19	82	207	284	244	109	16	1	0	(11)
(12)		CDH23.3	13631	0	2	27	268	987	2756	4203	3454	1579	331	25	0	(12)
(13)		CDH26.7	7420	0	0	2	66	382	1517	2559	2048	749	94	4	0	(13)
(14)	Wind	WSAvg	1.6	1.5	1.9	2.2	1.8	1.7	2.0	1.7	1.4	1.3	1.2	1.1	1.4	(14)
(15)	Precipitation	PrecAvg	671	78	92	135	115	82	7	2	0	3	39	53	71	(15)
(16)		PrecMax	931	159	177	250	224	202	40	39	5	35	169	126	195	(16)
(17)		PrecMin	411	10	12	46	38	2	0	0	0	0	5	11		(17)
(18)		PrecStd	135	33	45	49	46	52	9	7	1	8	40	37	38	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	20.9	26.2	31.1	34.5	38.2	40.6	39.8	35.8	31.9	26.9	18.9	(19)
(20)			MCWB	8.1	9.7	13.2	17.0	17.9	19.1	20.3	19.8	16.4	14.4	12.5	9.0	(20)
(21)		2%	DB	14.8	16.9	23.8	28.5	32.2	36.9	39.1	37.9	34.2	29.1	22.4	15.9	(21)
(22)			MCWB	7.1	9.3	12.9	15.9	17.7	18.4	20.3	19.0	16.2	14.2	11.6	7.6	(22)
(23)		5%	DB	12.3	14.5	21.3	26.5	30.8	35.6	37.8	36.6	33.0	27.1	20.0	13.8	(23)
(24)			MCWB	6.2	7.6	11.7	15.6	17.4	18.4	19.3	18.3	16.2	13.9	10.6	6.9	(24)
(25)		10%	DB	10.2	12.8	19.1	24.2	28.9	34.0	36.2	35.1	31.1	24.8	17.6	11.5	(25)
(26)			MCWB	5.4	6.8	10.7	14.5	17.1	18.1	19.1	17.8	15.8	13.2	9.7	6.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	11.5	15.3	18.8	20.4	22.5	25.8	23.2	19.9	16.7	14.0	10.3	(27)
(28)			MCD B	14.7	18.0	23.6	26.8	29.8	33.9	35.9	35.5	32.1	26.8	23.5	15.7	(28)
(29)		2%	WB	8.1	9.7	13.7	17.4	19.2	21.1	23.4	21.5	18.3	15.4	12.6	8.7	(29)
(30)			MCD B	12.0	14.9	21.2	25.8	28.6	32.1	33.4	33.3	30.3	25.5	20.3	13.2	(30)
(31)		5%	WB	7.1	8.7	12.5	16.4	18.3	20.1	21.9	20.3	17.1	14.5	11.4	7.8	(31)
(32)			MCD B	10.8	13.0	19.1	24.2	27.6	31.1	33.4	32.3	29.2	24.7	17.8	11.5	(32)
(33)		10%	WB	6.1	7.6	11.5	15.4	17.6	19.2	20.7	19.2	16.2	13.6	10.4	6.7	(33)
(34)			MCD B	9.3	11.3	17.4	22.3	26.5	30.4	32.5	31.6	28.6	23.1	15.9	10.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.6	9.6	10.8	12.1	13.7	15.8	17.2	17.9	17.9	15.8	12.0	10.3	(35)
(36)			MCD BR	12.8	12.8	14.4	16.4	17.5	18.4	19.5	19.6	20.4	19.1	16.8	14.3	(36)
(37)			MCWBR	7.1	6.9	7.1	7.4	6.9	6.6	6.8	6.7	7.4	7.6	7.8	7.7	(37)
(38)		5% WB	MCD BR	10.4	10.5	12.2	13.8	14.8	16.4	17.0	17.9	18.5	16.5	13.6	11.1	(38)
(39)			MCWBR	6.3	6.2	6.6	6.9	6.5	6.7	7.3	7.2	7.3	7.1	7.0	6.7	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.369	0.421	0.492	0.505	0.517	0.547	0.546	0.504	0.464	0.485	0.432	0.374	(41)	
(42)		taud	2.098	1.964	1.839	1.866	1.880	1.814	1.811	1.898	1.955	1.865	1.960	2.067	(42)	
(43)		Ebn at Noon	770	774	758	780	779	755	750	771	775	695	682	725	(43)	
		Edh at Noon	116	152	191	197	198	212	211	188	167	166	133	112		
(44)	All-Sky Solar Radiation	RadAvg	2.05	2.67	3.78	5.13	6.37	7.52	7.67	7.12	5.89	3.98	2.38	1.82	(44)	
(45)		RadStd	0.26	0.44	0.38	0.57	0.50	0.36	0.25	0.28	0.21	0.39	0.32	0.25	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	+0.44	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page